

FIITJEE

ALL INDIA TEST SERIES

FULL TEST – VI

JEE (Main)-2024

TEST DATE: 18-01-2024

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1. B
Sol. Conceptual

2. C
Sol. i_0 is given as the main current. Current within shaded are

$$i = \frac{i_0 \pi r^2}{\pi R^2} = i_0 \frac{r^2}{R^2}$$

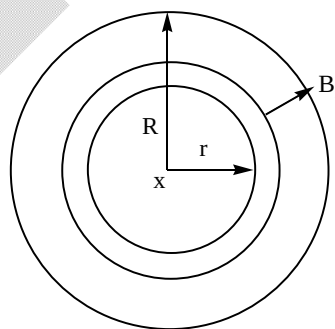
By Ampere's law

$$B_r 2\pi r = \mu_0 i = \mu_0 i_0 \frac{r^2}{R^2}$$

$$B_r = \frac{\mu_0 i_0}{2\pi R^2} r$$

$$U = \text{Energy stored} = \int_0^R \left(\frac{B_r^2}{2\mu_0} \right) (2\pi r dr) \ell$$

$$\Rightarrow \int_0^R \frac{\mu_0^2 i_0^2 r^2}{4\pi^2 R^4} \frac{2\pi r dr}{2\mu_0} \ell = \frac{\mu_0 i_0^2}{16\pi} \ell$$



3. A

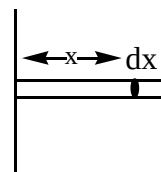
 Sol. Change in length of element dx due to temperature difference.

$$du = (dx)\alpha \Delta T$$

$$du = (dx)\alpha \frac{T_0 x^2}{L^2}$$

$$\int du = \int_0^L \frac{\alpha T_0 x^2}{L^2} dx \Rightarrow u = \frac{\alpha T_0 L^3}{3L^2}$$

$$\text{Stress} = E \times \frac{\Delta L}{L} = \frac{\alpha T_0}{3L^2} \frac{L^3}{L} \cdot E = \frac{E\alpha T_0}{3}$$



4. B

 Sol. $x = A \cos \omega t$

$$\frac{4A}{5} = A \cos \omega t \Rightarrow \cos \omega t = \frac{4}{5}$$

$$\omega t = \cos^{-1}\left(\frac{4}{5}\right)$$

$$t = \frac{1}{\omega} \cos^{-1}\left(\frac{4}{5}\right) = \frac{T}{2\pi} \cos^{-1}\left(\frac{4}{5}\right) \text{ s}$$

5. B

 Sol. Intensity at a distance x from point source is given by $I = \frac{P}{4\pi x^2}$

 Consider on elemental shell of radius x and thickness dx . In this region, the energy contained is

$$dE = \frac{I}{c} 4\pi x^2 dx = \frac{P dx}{c}$$

 Let 'dn' be the number of photons in this elemental shell then $dn \times \frac{hc}{\lambda} = dE = \frac{P dx}{c}$

$$dn = \frac{P \lambda dx}{hc^2}$$

 Total number of photons in the shell of inner radius 'r' and outer radius $2r$ is

$$n = \int dn = \int_r^{2r} \frac{P \lambda dx}{hc^2} = \frac{P \lambda r}{hc^2}$$

6. C

Sol. Law of conservation of angular momentum about point of contact

$$mv_0 R = mvR + I\omega$$

$$mv_0 R = mvR + \frac{mR^2}{2} \left(\frac{v}{R}\right)$$

$$v = \frac{2}{3} v_0$$

Workdone by friction = loss of K E

$$\begin{aligned}
 &= \frac{1}{2}mv_0^2 - \left(\frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 \right) \\
 &= \frac{1}{2}mV_0^2 - \left[\frac{3}{4}m \left(\frac{2}{3}v_0 \right)^2 \right] \\
 &= \frac{1}{6}mv_0^2
 \end{aligned}$$

7. A

Sol. $\mu mg = ma$

$$F - \mu mg = M\mu g$$

$$F = \mu g(M + m) = 6N$$

8. A

Sol. Suppose initial current is i_0

$$i(t) = i_0 \left(1 - \frac{t}{\Delta t} \right)$$

$$q = i_0 \int_0^{\Delta t} \left(1 - \frac{t}{\Delta t} \right) dt$$

$$\text{So } i_0 = \frac{2q}{\Delta t}$$

$$H = \int_0^{\Delta t} \left(\frac{2q}{\Delta t} \left(1 - \frac{t}{\Delta t} \right) \right)^2 R dt = \frac{4}{3} \frac{q^2 R}{\Delta t}$$

9. A

Sol. $V_C = 2V_R$

$$\frac{q}{C} = 2IR$$

$$\frac{CE \left(1 - e^{-\frac{t}{RC}} \right)}{C} = 2 \cdot \frac{E}{R} e^{-\frac{t}{RC}R}$$

$$T = RC \log_e 3$$

10. B

Sol. $F = q[\vec{E} + \vec{V} \times \vec{B}]$

$$= q[(3\vec{i} + \vec{j} + 2\vec{k}) + (3\vec{i} + 4\vec{j} + \vec{k}) \times (\vec{i} + \vec{j} - 3\vec{k})]$$

$$= q[-10\vec{i} + 11\vec{j} + \vec{k}]$$

$$F_y = 11q$$

11. A

Sol. Conserve linear momentum in horizontal direction and apply law of conservation of energy for translatory motion

12. C

$$\text{Sol. } P = t^1 = \frac{V_0}{g} = \frac{at}{g}$$

$$(a_t)V = k^2 r^2 t$$

$$V \frac{dv}{dt} = k^2 r^2 t$$

$$\frac{V^2}{2} = \frac{k^2 r^2 t^2}{2} \quad \text{or } V = krt$$

$$a_t = \frac{dv}{dt} = kr$$

$$a_c = \frac{V^2}{r} = \frac{k^2 r^2 t^2}{r} = k^2 r t^2$$

13. C

$$\text{Sol. } a = \frac{f}{M}$$

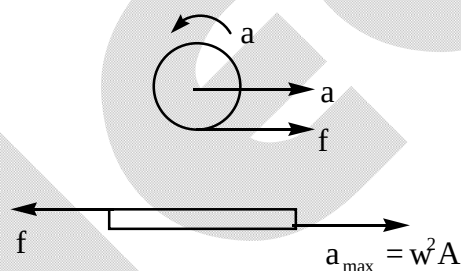
$$\alpha = \frac{f(R)}{MR^2} = \frac{2f}{MR}$$

$$a + R\alpha = a_{\max} = \omega^2 A$$

$$\frac{3f}{M} = \omega^2 A$$

$$f = \frac{M\omega^2 A}{3}$$

$$\tau_{\max} = \frac{1}{3} M\omega^2 AR$$



14. D

$$\text{Sol. Acceleration when block B is in liquid } a_1 = \frac{m_A g - (m_B g - \text{up thrust})}{m_A + 3m}$$

$$= \frac{m_A g - \left(3mg - \frac{3m}{2\rho} \rho g \right)}{m_A + 3m} (\uparrow)$$

$$\text{Acceleration when block B is outside the liquid is } a_2 = \frac{3mg - m_A g}{m_A + 3m} (\downarrow)$$

$$\text{Given } a_1 = a_2$$

$$m_A g - \frac{3}{2} mg = 3mg - m_A g$$

$$2m_A g = \frac{9}{2} mg$$

$$m_A = \left(\frac{9}{4} \right) m$$

15. B

Sol. $\Delta Q = nC(\Delta T)$

$$C = \frac{R}{r-1} + \frac{R}{1-x} = \frac{R}{\frac{5}{3}-1} + \frac{R}{1-x} = \frac{3R}{2} + \frac{R}{1-x}$$

$$\Delta Q = n \left[\frac{3R}{2} + \frac{R}{1-x} \right] (2T_0 - T_0) = (1)(R)(T_0) \left[\frac{5-3x}{2(1-x)} \right] = 2RT_0$$

$$X = -1$$

$$PV^{-1} = \text{constant}$$

16. A

Sol. Conceptual

17. D

$$\text{Sol. } \vec{\tau} = \vec{M} \times \vec{B} = (4\vec{i} - 3\vec{j}) \times (3\vec{i} + 4\vec{j}) = 25\vec{k} \text{ (N-m)}$$

Moment of inertia about an axis passing through centre of mass and parallel to $\vec{\tau}$ is

$$I = \frac{10^{-2}}{2}$$

$$\alpha = \frac{\tau}{I} = \frac{25}{0.5 \times 10^{-2}} = 5000 \text{ rad/sec}^2$$

$$\frac{1}{2} I \omega^2 = U_i - U_f = -MB \cos 90^\circ - (-MB \cos 0)$$

$$\omega = \sqrt{\frac{2MB}{I}} = 100 \text{ rad/sec}$$

18. A

$$\text{Sol. } a_x = \frac{0.9}{0.5} = 1.8 \text{ ms}^{-2}$$

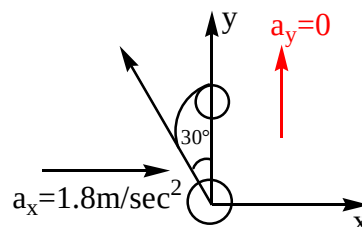
$$a_y = 0$$

When the sphere crosses the y-axis, the displacement component along x-axis is zero.

$$0 = u_x t + \frac{1}{2} a_x t^2$$

$$0 = (3 \sin 30^\circ) t - \frac{1}{2} (1.8) t^2$$

$$T = 1.66 \text{ sec}$$



19. D

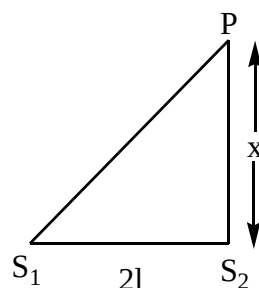
$$\text{Sol. } \lambda_m \text{ decreases on increase of } V \left(\lambda_m = \frac{12400}{V} (\text{\AA}) \right)$$

Energy of emitted photons will increase as λ_m decreases. Intensity is energy per unit area per unit time. Hence intensity will increase

20. C

$$\text{Sol. } (4\lambda^2 + x^2)^{1/2} - x = \frac{3\lambda}{2}$$

$$x = \frac{7\lambda}{12}$$



SECTION – B

21. 4

$$\text{Sol. } r_{\min} = 1.8 \times 10^{12} \text{ m, } a = 2 \times 10^{12} \text{ m}$$

$$r_{\max} = 2a - r = 2.2 \times 10^{12} \text{ m}$$

$$V_{\max} \times r_{\min} = V_{\min} \times r_{\max}$$

$$\frac{dA}{dt} = \frac{V_{\max} \times r_{\min}}{2} = \frac{V_{\min} \times r_{\max}}{2}$$

$$V_{\min} = 40 \text{ km / sec}$$

$$X = 4$$

22. 1

$$\text{Sol. For centre of face } \phi_1 = \frac{1}{2} \frac{q}{\epsilon_0} = \frac{q}{4\epsilon_0}$$

$$\text{For corner of a cube } \phi_2 = \frac{1}{8} \frac{2q}{\epsilon_0} = \frac{q}{4\epsilon_0}$$

$$\text{Total flux } \phi = \frac{q}{4\epsilon_0} + \frac{q}{4\epsilon_0} = \frac{q}{2\epsilon_0}$$

23. 8

$$\text{Sol. } t = \frac{t_1 t_2}{t_1 + t_2} = \frac{6 \times 3}{9} = 2 \text{ hr}$$

6 hr. is equal to three half lines

$$N = \frac{N_0}{2^3} = \frac{N_0}{8} \Rightarrow \frac{N_0}{N} = 8$$

24. 1

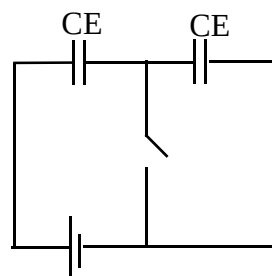
Sol. Initially both the capacitors has a charge of CE as shown when switch is closed capacitor 2 starts discharging and capacitor '1' will get final charge 2CE

So, battery supplies CE more charge

$$\text{Work done by battery} = (CE) (2E) = 2CE^2$$

$$U_i = \frac{CE^2}{2C} + \frac{CE^2}{2C} = CE^2$$

$$U_f = \frac{(2CE)^2}{2C} = 2CE^2$$



From energy conservation $U_i + W_B = U_f + \Delta H$

$$\Delta H = CE^2$$

25. 8

Sol. $dF = (\rho gy)(4dy)$

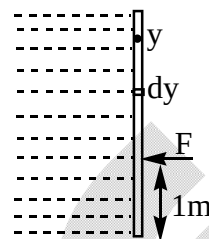
Torque about hinge

$$d\tau = (\rho gy)(4dy)(y)$$

$$\tau = \int_0^4 4\rho gy^2 dy = 4\rho g \left(\frac{y^3}{3} \right)_0^4$$

$$\tau = F(3)$$

$$(4\rho g) \left(\frac{4^3}{3} \right) = F(3) \quad F = \frac{2^8 \rho g}{9} \quad n = 8$$



26. 1

Sol. $y = Ae^{-(5x^2 - 40x + 80)} = Ae^{-5(x-4)^2}$

$$y = Ae^{-b(kx - \omega t)^2}$$

$$k = 1, \omega = 2 \quad V = \frac{\omega}{k} = 2 \text{ m/sec}$$

27. 29

Sol. $Z - 1 = \sqrt{\frac{4v}{3cR}}$

$$Z - 1 \propto \sqrt{\frac{1}{\lambda}}$$

$$\frac{Z_2 - 1}{Z_1 - 1} = \sqrt{\frac{\lambda_1}{\lambda_2}} = \sqrt{\frac{196}{169}} = \frac{14}{13}$$

$$Z_2 - 1 = 26 \times \frac{14}{13} = 28$$

$$Z_2 = 29$$

28. 2

Sol. Rate of radiation $= \sigma eAT^4 = \text{constant}$

$$\text{Or } r^2 T^4 = \text{constant}$$

$$\text{Or } \frac{4}{3} \pi r^3 T^6 = \text{constant}$$

$$\text{Or } VT^6 = C$$

$$\text{Or } TV^{1/6} = \text{constant} \quad \dots(i)$$

Comparing it with polytropic equation $TV^{K-1} = \text{constant}$

$$\text{We get } K - 1 = \frac{1}{6}$$

$$\Rightarrow K = \frac{7}{6}$$

$$\text{Work done by gas} = \frac{nR(T_1 - T_2)}{K - 1} = \frac{nR}{(1/6)}(T_0 - T_2)$$

$$\Rightarrow n = 2$$

29. 7

Sol. At the instant $i = \frac{-dq}{dt} = \sqrt{3} \text{ c / sec}$

$$\frac{q_{\max}^2}{2C} = \frac{q^2}{2C} + \frac{1}{2}Li^2$$

$$q_{\max} = (q^2 + LCi^2)^{1/2} = 7$$

30. 4

Sol. Dots will be aligned again after time 't', if slower disc makes one rotation less than the other disc

$$2\pi n = \frac{1}{2} \left(\frac{\pi}{2} \right) t^2 \quad \dots (i)$$

$$2\pi(n-1) = \frac{1}{2} \left(\frac{\pi}{4} \right) t^2 \quad \dots (ii)$$

$$\frac{n}{n-1} = \frac{2}{1} \Rightarrow n = 2$$

From equation (i) $t = 4 \text{ sec}$

Chemistry

PART – B

SECTION – A

31. D

Sol. Metal with d^8 configuration is ideal for the formation of low spin square planar complexes with strong field ligands. Tetrahedral high spin complexes are typically formed with bulky ligands due to less steric hinderance than the square planar complex. The importance of electronic and steric balance is apparent in complexes such as $[\text{Ni}(\text{PPh}_2\text{Et})_2\text{Br}_2]$ which can be isolated as a green tetrahedral complex and also as a brown square planar complex.

(C) All the complexes with Ni^{2+} (d^8 configuration) are paramagnetic owing to 2 unpaired electrons.

32. A

Sol. Zr and Hf belong to transition series. The oxidising power of Mn in MnO_4^- is highly pH dependent.

33. B

Sol. $\text{Na}_2\text{H}_2\text{P}_2\text{O}_7 + 2\text{NaHCO}_3 \longrightarrow \text{Na}_4\text{P}_2\text{O}_7 + 2\text{CO}_2 + 2\text{H}_2\text{O}$

34. C

Sol. $\text{KClO}_3 \longrightarrow \text{KCl} + \text{O}_2$

$\text{HOCl} \longrightarrow \text{HCl} + \text{O}_2$

$\text{LiNO}_3 \longrightarrow \text{Li}_2\text{O} + \text{NO}_2 + \text{O}_2$

$\text{NaHCO}_3 \longrightarrow \text{Na}_2\text{CO}_3 + \text{CO}_2 + \text{H}_2\text{O}$

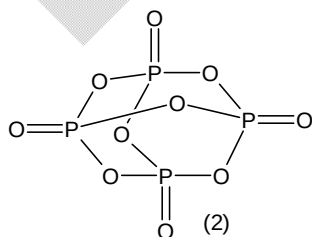
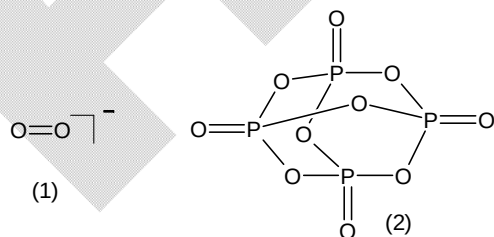
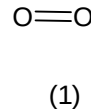
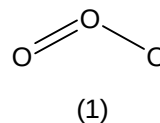
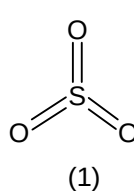
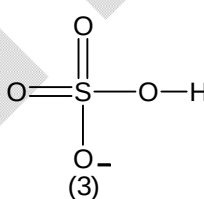
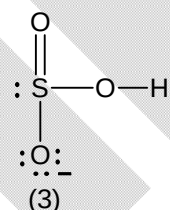
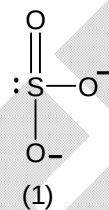
$\text{CO} + \text{H}_2\text{O}(\text{steam}) \longrightarrow \text{CO}_2 + \text{H}_2$

$\text{NH}_4\text{NO}_3 \longrightarrow \text{N}_2\text{O} + \text{H}_2\text{O}$

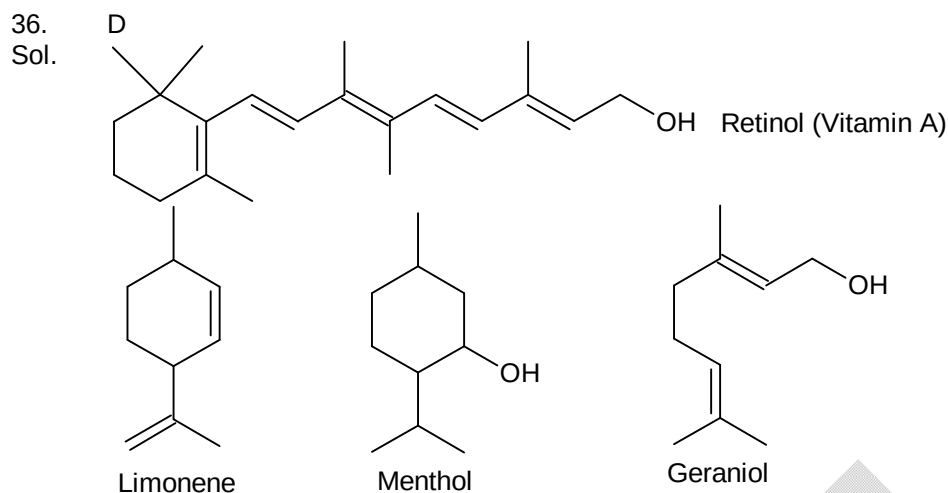
$\text{NH}_4\text{NO}_2 \longrightarrow \text{N}_2 + \text{H}_2\text{O}$

35. C

Sol.

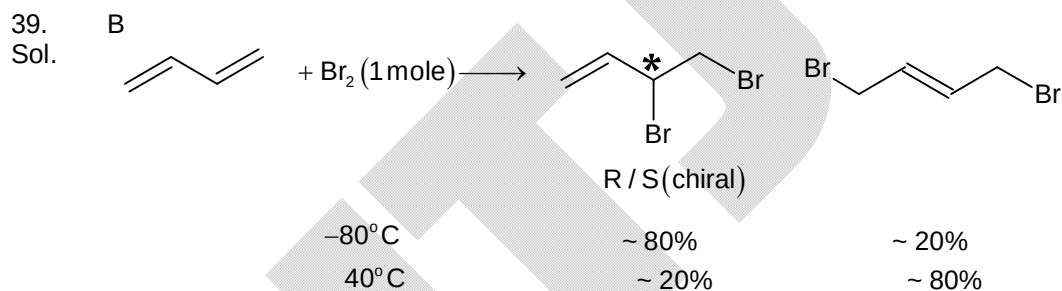


Total = 13



37. A
Sol. The triplet carbene with the two unbonded electrons are present in two degenerate p-orbitals, having same spin brings greater stability to this electrons configuration.

38. A
Sol. 1, 6, 7-Trimethylnaphthalene, with lowest sets of locants preferred over 2, 3, 5-Trimethylnaphthalene, with lowest sum of locants.



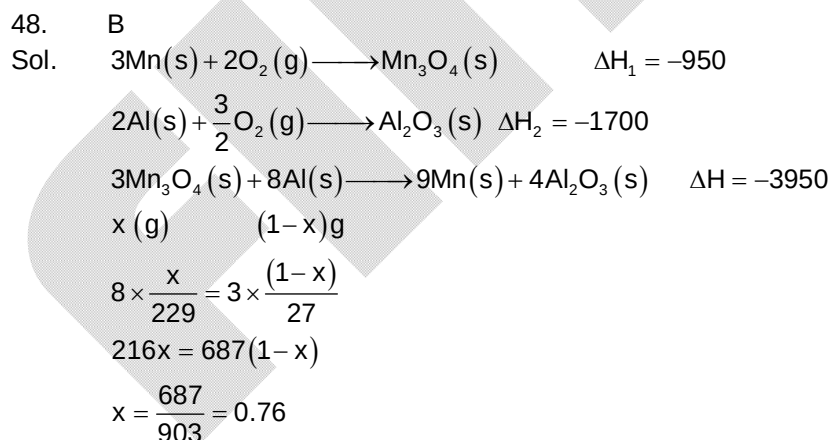
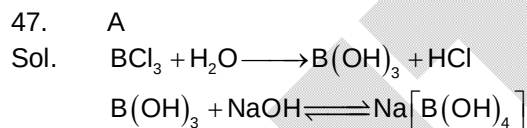
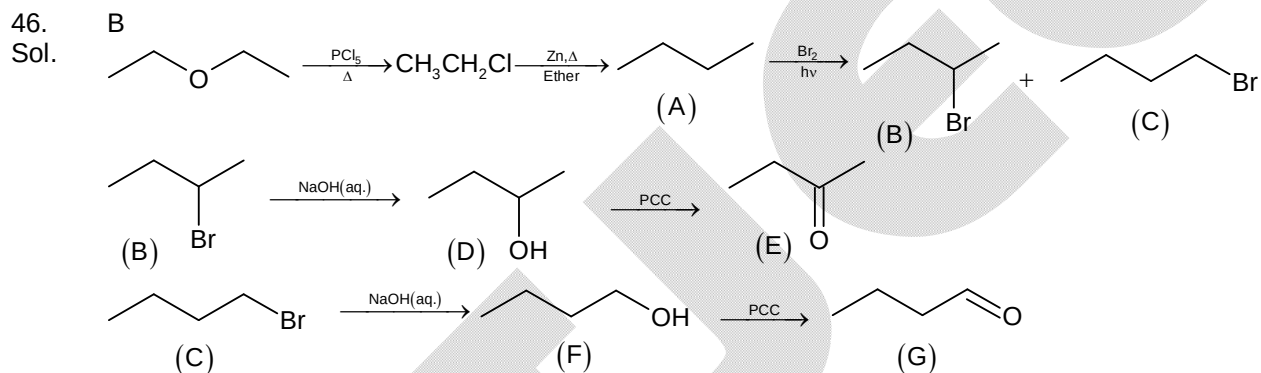
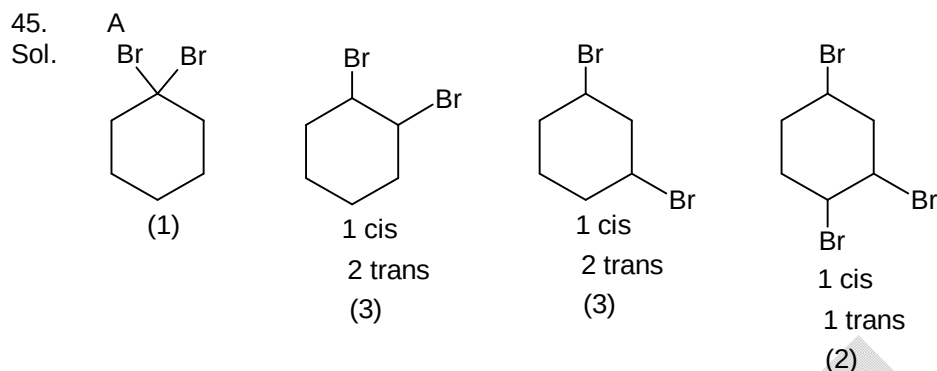
40. A
Sol. H_2 reacts over 13 times faster with Cl_2 than D_2 , because H_2 has a lower energy of activation.

41. D
Sol. Difference in position of attack of electrophile on substituted benzene reflects the relative electron-density of the various positions.

42. C
Sol. Possible reaction at cathode $\text{Cu}^{++} + 2\text{e} \rightarrow \text{Cu}$; $E^\circ = +0.34\text{V}$. [$\Delta G = \text{negative}$]
 $\text{NO}_3^- + 2\text{H}^+ + \text{e} \rightarrow \text{NO}_2 + \text{H}_2\text{O}$; $E^\circ = -0.40\text{V}$ [$\Delta G = \text{positive}$]
So copper is deposited at cathode
At anode: $2\text{H}_2\text{O} \longrightarrow 4\text{H}^+ + \text{O}_2 + 4\text{e}$; so oxygen is evolved at anode.

43. D
Sol. $\text{A}_x\text{B}_y \rightleftharpoons x\text{A}^{y+} + y\text{B}^{x-}$
 $\text{C}(1-\alpha) \quad x\alpha\text{C} \quad y\alpha\text{C}$
 $K_{\text{eq}} = x^x y^y \alpha^{(x+y)} \text{C}^{(x+y-1)}$

44. B
Sol. D-from the first solution and OD from the second solution attach on the same side of the double bond.



Required energy released on reduction of $\frac{0.76}{229}$ mole of Mn_3O_4 (in 1 g of the mixture) = 4.37 kJ

49. B
Sol. Mn shows violet colour in oxidising flame during borax bead test.

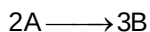
50. D

Sol. Reaction of phenol with bromine water involve electrophilic substitution step.

SECTION – B

51. 6

Sol.



$$\text{time} = 0 \quad A_0 \quad 0$$

$$-2x \quad 3x$$

$$\text{time} = t \quad A_0 - 2x \quad 3x$$

At point of intersection,

$$A_0 - 2x = 3x$$

$$A_0 = 5x$$

$$\frac{A_0}{5} = x$$

$$[B] = \frac{3}{5} A_0$$

$$N = \frac{3}{5}$$

52. 150

$$\text{Sol. } \Delta x \cdot m \cdot \Delta V \geq \frac{h}{4\pi}$$

$$T = \frac{300}{2}$$

$$m = \frac{h}{4\pi \Delta x \cdot \Delta V}$$

$$m = \frac{6.62 \times 10^{-34}}{4 \times 3.14 \times 4.960 \times 10^{-10}} \text{ kg}$$

$$m = 0.106 \times 10^{-24} \text{ kg}$$

$$m = 0.106 \times 10^{-24} \times 6.023 \times 10^{23} \times 10^3 \text{ g mol}^{-1}$$

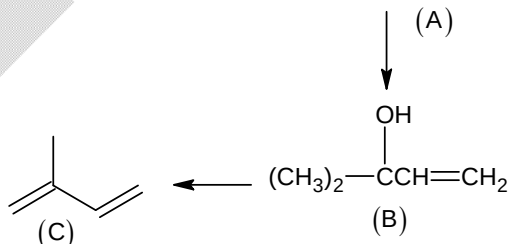
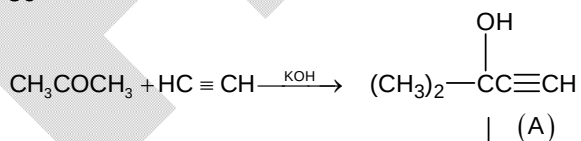
$$m = 64 \text{ g mol}^{-1}$$

$$\sqrt{\frac{3RT}{M}} = \sqrt{\frac{3RT'}{M'}}$$

$$\frac{300}{64} = \frac{T'}{32}$$

53. 86

Sol.



54. 39

Sol. $\text{CaCO}_3(\text{s}) \rightleftharpoons \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$, $K_p = 6.15 \text{ atm}$

CaCO_3 first appears once the reaction achieve equilibrium, means partial pressure inside the vessel reaches 6.15 atm.

$$\text{So, volume will be} = \frac{1 \times 0.082 \times 300}{6.15} = 4 \text{ litre}$$

$$w = -nRT \ln \frac{V_f}{V_i} = -0.082 \times 300 \times \ln \frac{4}{20}$$

$$w = 0.082 \times 300 \ln 5 = 39.36 \text{ J}$$

55. 4

Sol. First AgI will precipitate upon addition of AgNO_3 . When AgCl starts precipitating.

$$[\text{Ag}^+][\text{Cl}^-] = K_{sp}(\text{AgCl})$$

$$[\text{Ag}^+] \times 0.01 = 2 \times 10^{-10}$$

$$[\text{Ag}^+] = 2 \times 10^{-8}$$

$$[\text{Ag}^+][\text{I}^-] = K_{sp}(\text{AgI})$$

$$2 \times 10^{-8} \times [\text{I}^-] = 8 \times 10^{-17}$$

$$[\text{I}^-] = \frac{8 \times 10^{-17}}{2 \times 10^{-8}}$$

$$[\text{I}^-] = 4 \times 10^{-9} \text{ M}$$

56. 20

Sol. To bring down the depression in freezing point 1.1625°C . Molality of the aq. solution is

$$\Delta T_f = K_f m$$

$$1.1625 = 1.86 \times \frac{1 \text{ mole}}{\text{mass of water (in kg)}}$$

$$\text{Mass of remaining water} = 1.6 \text{ kg}$$

$$\text{Amount of water removed} = 2 - 1.6 = 0.4 \text{ kg}$$

$$\text{Required \%} = \frac{0.4}{2} \times 100 = 20\%$$

57. 676

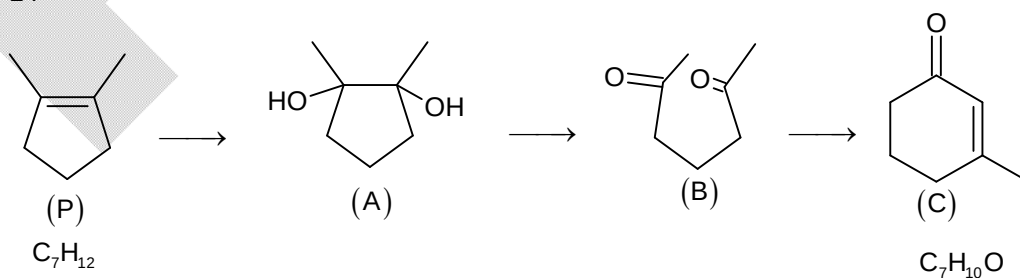
Sol. (i) $3\text{I}^- \longrightarrow \text{I}_3^- + 2\text{e}^-$ $E_1^\circ = -0.5355 \text{ V}$ (ii) $\text{I}_2 + 2\text{e}^- \longrightarrow 2\text{I}^-$ $E_2^\circ = 0.619 \text{ V}$ (iii) $\text{I}_2 + \text{I}^- \longrightarrow \text{I}_3^-$ $E_1^\circ + E_2^\circ$

$$\Delta G = \Delta G^\circ + RT \ln Q$$

$$\Delta G^\circ = -RT \ln K_{eq}$$

58. 14

Sol.



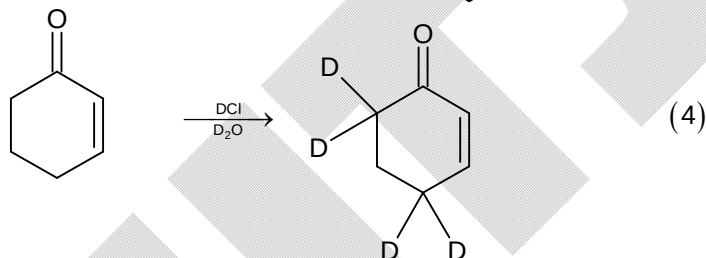
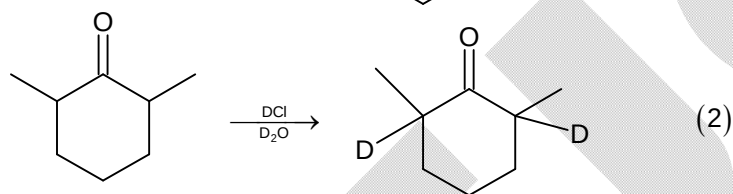
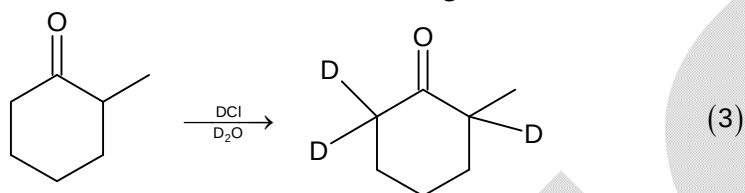
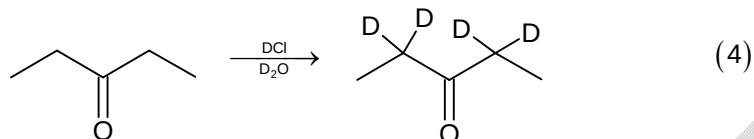
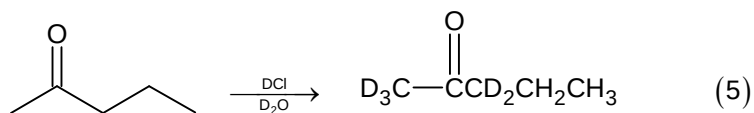
$$10 + 16 - 12 = 26 - 12 = 14$$

59. 4

Sol. The main chain in amylopectin consists of an α -1,4-glycosidic linkage. Amylose is the water insoluble component of starch. Glucose phenylhydrazine is soluble in water.

60. 18

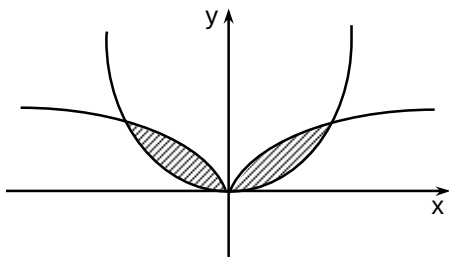
Sol.



Mathematics**PART – C****SECTION – A**

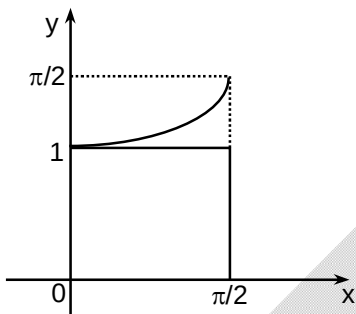
61. A

Sol. $\left(\frac{16}{3}ab\right) \times 2$



62. A

Sol. $\frac{x}{\sin x} = F(x)$



63. B

Sol. $T_r = \tan^{-1}\left(r + \frac{1}{2}\right) - \tan^{-1}\left(r - \frac{1}{2}\right)$

64. A

Sol. $x_i = 1, y_i = -1.$

65. C

Sol. $I = \frac{1}{2007} \ln\left(\frac{x^{2007}}{1+x^{2007}}\right).$

66. A

Sol. $S = 44\frac{1}{2}$

67. A

Sol. $\vec{r} \cdot (\vec{a} + \vec{b} + \vec{c}) = 0.$

68. C

Sol. Let $\vec{a}, \vec{b}, \vec{c}$ be the position vector of A, B, C respectively

$\therefore 2\vec{a} + 5\vec{b} + 10\vec{c} = \vec{0}.$

69. C
Sol. Equation of line through $O(0, 0, 0)$ and perpendicular to the plane $2x - y - z = 4$, is

$$\frac{x-0}{2} = \frac{y-0}{-1} = \frac{z-0}{-1}$$

70. D
Sol. $a = 0$

$$x + y = \frac{\pi}{4}$$

71. D
Sol. $N = 2^2 \cdot 3^5 \cdot 5^5$.

72. A
Sol. Reflexive, transitive but not symmetric.

73. B
Sol. $|A^n - I| = 1 - 2^n$.

74. A
Sol. $m = 123$

75. A
Sol. $(z^2 + 1)(z^2 + z + 1) = z^2$
 $\Rightarrow z^5 = 1 \Rightarrow z^{100} = 1$.

76. B
Sol. $|z - \sqrt{2}(1+i)| = |z - 0|$.

77. D
Sol. $m = \pi$.

78. B
Sol. $(1+z)^3$ where $z = x(1+2x+3x^2)$
 $(1+y)^3$, where $y = x(1+2x+3x^2+4x^3)$

79. D
Sol. $2a^2 + 8b^2 + 18c^2 + 4ab + 12bc + 6ac \leq 0$.
 $\Rightarrow (a+2b)^2 + (2b+3c)^2 + (3c+a)^2 \leq 0$
 $a = b = c = 0$

80. A
Sol. $f(x) + f(-x) = 8$

SECTION - B

81. 11
Sol. Median = $\frac{2k+12}{2} = k+6$

$$\text{Mean deviation} = 6 \Rightarrow \frac{58-2k}{8} = 6$$

$$\Rightarrow k = 5$$

82. 5

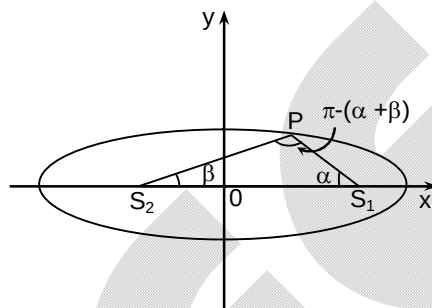
Sol. $lx + my + n = 0$ and the point $\left(\frac{t^3}{t-1}, \frac{t^2-3}{t-1}\right)$ lie on the line where $t = a, b, c$.

83. 2

Sol. $(0, 2)$

84. 2

Sol. $\frac{2ae}{\sin(\alpha + \beta)} = \frac{S_1P}{\sin\beta} = \frac{S_2P}{\sin\alpha} = \frac{2a}{\sin\alpha + \sin\beta}$.

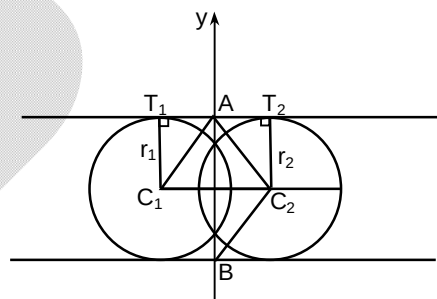


85. 2

Sol. $d_n = \left| \frac{n - \sqrt{n^2 + 1}}{\sqrt{2}} \right|$.

86. 95

Sol. $AT_1 = AT_2$
 $C_1A^2 = C_1T_1^2 + AT_1^2$
 $C_2B^2 = C_2A^2 = C_2T_2^2 + AT_2^2$
 $C_1A^2 - C_2B^2 = r_1^2 - r_2^2 = \frac{91}{4}$.



87. 2

Sol. $k = 2$.

88. 3

Sol. $P(A \cap B) = P(A)P(B)$

89. 3

Sol. $\frac{3}{\sqrt{2}}$

90. 5

Sol. $(x - 3)^2 + (y - 0)^2 = 5$